is a type of convergence series

Adv

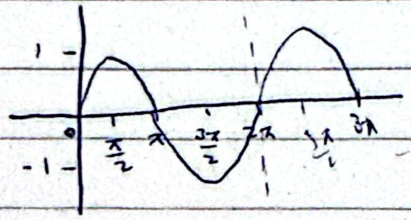
FOURIER SERIES [Chap 11] 11.1, 11.2

(i) function must be periodic of period 2π

example $\sin(x+2\pi) = \sin x$

$$f(x+2\pi) = f(x)$$

All trignometric functions are periodic with a period of 2π



If $f(x)$ is a periodic function of period 2π then its fourier series is defined as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx.$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

Euler's Coefficients

$-\pi$ to π = 2π → Length

-12-21

$x \to \sin \to 0$
$y \to \cos \to 0$

Only 2 can be 0 together

All 3 a_0, a_n and b_n cannot be 0 together

Q) Find the Fourier Series for the following function.

(i) $f(x) = \begin{cases} -1 & ; -\pi < x < 0 \\ 1 & \;\; 0 < x < \pi \end{cases}$; $\underbrace{f(x) = f(x+2\pi)}$.

This indicates periodicity of period 2π aka cyclic function

Required fourier series is:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0} (-1)\,dx + \int_{0}^{\pi} (1)\,dx\right]$$

$$= \frac{1}{2\pi}\left[-x \Big|_{-\pi}^{0} + x \Big|_{0}^{\pi} \right]$$

$$\cancel{= \frac{1}{2\pi}\left[(-(-\pi) - 0) + (\pi - 0)\right]}$$

$$= \frac{1}{2\pi}\left[-(-\pi - 0) + (\pi - 0)\right] = 0.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} (-1) \cos nx + \int_{0}^{\pi} (1) \cos nx\right]$$

$$= \frac{1}{\pi}\left[-\left\{\frac{\sin nx}{n}\Big|_{-\pi}^{0}\right\} + \left\{\frac{\sin x}{n}\Big|_{0}^{\pi}\right\}\right]$$

$$= \frac{1}{\pi}\left[-\frac{1}{n}\left\{\sin(0) - \sin n(-\pi)\right\} + \frac{1}{n}\left\{\sin n\pi - \sin 0\right\}\right]$$

$$= \frac{1}{\pi}\left[0 + 0\right] = 0.$$

indicates periodicity of period 2π

aka cyclic function

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0}(-1)\,dx + \int_{0}^{\pi}(1)\,dx\right]$$

$$= \frac{1}{2\pi}\left[\left(-x\Big|_{-\pi}^{0}\right) + x\Big|_{0}^{\pi}\right]$$

$$\cancel{= \frac{1}{2\pi}\left[(-(-\pi)-0) + (\pi-0)\right]}$$

$$= \frac{1}{2\pi}\left[-(-\pi-0) + (\pi-0)\right] = 0.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}(-1)\cos nx + \int_{0}^{\pi}(1)\,\cos nx\right]$$

$$= \frac{1}{\pi}\left[-\left\{\frac{\sin nx}{n}\Big|_{-\pi}^{0}\right\} + \left\{\frac{\sin x}{n}\Big|_{0}^{\pi}\right\}\right]$$

$$= \frac{1}{\pi}\left[-\frac{1}{n}\left\{\sin(0) - \sin n(-\pi)\right\} + \frac{1}{n}\left\{\sin n\pi - \sin 0\right\}\right]$$

$$= \frac{1}{\pi}\left[0+0\right] = 0.$$

so b_n cannot be zero

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}(-1)\sin nx\,dx + \int_{0}^{\pi}(1)\sin nx\,dx\right.$$

$$= \frac{1}{\pi}\left[\frac{\cos nx}{n}\Big|_{-\pi}^{0} + \left\{\frac{-\cos nx}{n}\Big|_{0}^{\pi}\right\}\right]$$

$$= \frac{1}{n\pi}\left[\{\cos 0 - \cos n(-\pi)\} - \{\cos n\pi - \cos 0\}\right]$$

$$= \frac{1}{n\pi}\left[\{1 - \cos n\pi\} - \{\cos n\pi - 1\}\right.$$

$$= \frac{1}{n\pi}[2 - 2\cos n\pi]$$

$$= \frac{2}{n\pi}[1-(-1)^n]$$

Putting all these coefficients in the eq.

$$\text{eq } 1 \Rightarrow f(x) = 0 + \sum_{n=1}^{\infty}(0)\cos nx + \frac{2}{n\pi}\{1-(-1)^n\}\sin nx$$

$$\begin{cases} -1 & -\pi<x<0 \\ 1 & 0<x<\pi \end{cases} = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\{1-(-1)^n\}}{n}\sin nx$$

Ans

$$\frac{2}{\pi}\left[\frac{1-(-1)^1}{1}\sin x + \frac{1-(-1)^2}{2}\sin 2x + \frac{1-(-1)^3}{3}\sin 3x\right.$$

$$\left. + \frac{1-(-1)^4}{4}\sin 4x + \frac{1-(-1)^5}{5}\sin 5x + \dots\right]$$

Odd adds, even becomes zero.

$$\begin{cases} -1 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases}$$

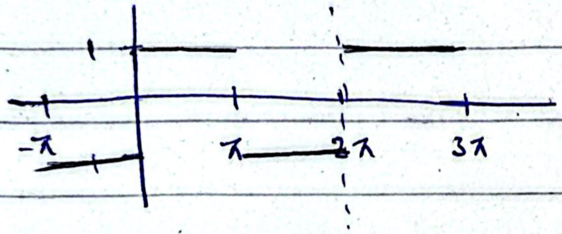
$$\boxed{\therefore f(x) = \frac{2}{\pi}\left[2\sin x + \frac{2\sin 3x}{3} + \frac{2\sin 5x}{5} + \frac{2\sin 7x}{7} + \dots\right]}$$

$$= \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\sin(2n-1)x}{(2n-1)}.$$

1, 3, 5, 7, 9

$T_n = a + (n-1)d$

$= 2n-1$



$$\frac{4}{\pi}\left[\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \frac{\sin 7x}{7} + \frac{\sin 9x}{9} + \dots\right]$$

let $x = \pi/2$

$$1 = \frac{4}{\pi}\left[\sin \pi/2 + \frac{\sin(3\pi/2)}{3} + \frac{\sin(5\pi/2)}{5} + \frac{\sin(7\pi/2)}{7} + \frac{\sin 9\pi/2}{9}\right.$$

$$1 = \frac{4}{\pi}\left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} + \dots\right]$$

$$\pi \cong 4\left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9}\right]$$

$\pi \cong 3.339$ → the further we go we can get the exact value of irrational number like π, $\sqrt{2}$, $\sqrt{3}$ etc.

* If $f(x)$ is an even function

$$\int_{-a}^{a} f = 2\int_{0}^{a} f$$

* If $f(x)$ is an odd functions

$$\int_{-a}^{a} f = 0$$

Ex. 11.1

Q_{12-21}

Q_{12} $f(x)$ in problem 6.

$f(x) = |x|$; $-\pi < x < \pi$ → not periodic?

or

$$f(x) = \begin{cases} -x & ; -\pi < x \le 0 \\ x & ; 0 < x \le \pi \end{cases}$$

Required F. Series:-

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{2\pi}\left[\int_{-\pi}^{0} -x\,dx + \int_{0}^{\pi} x\,dx\right]$$

$$= \frac{1}{2\pi}\left[-\left(\frac{x^2}{2}\Big|_{-\pi}^{0}\right) + \left(\frac{x^2}{2}\Big|_{0}^{\pi}\right)\right]$$

$$= \frac{1}{2\pi}\left[\left[0 + \frac{\pi^2}{2}\right] + \left[\frac{\pi^2}{2} + 0\right]\right]$$

$$\frac{1}{\cancel{2}\pi}\left(\frac{\cancel{2}\pi^{\cancel{2}}}{2}\right) = \frac{\pi}{2}$$

OR

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} |x|\,dx = \frac{1}{\cancel{2}\pi}\,\cancel{2}\int_{0}^{\pi} x\,dx$$

Even function → $f(-x) = f(x)$

Odd function → $f(-x) = -f(x)$

$$a_0 = \frac{1}{\pi}\left[\frac{x^2}{2}\Big|_0^{\pi}\right]$$

$$\frac{\pi^2}{2}\left(\frac{1}{\pi}\right) = \frac{\pi}{2}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx \, dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} |x| \cos nx \, dx$$

$$= \frac{1}{\pi}\, 2\int_{0}^{\pi} x\cos nx \, dx \qquad u = x \quad v' = \cos nx$$

$$u' = 1 \quad v = \frac{\sin nx}{n}$$

$$= \frac{1}{\pi}\, 2\left[x\,\frac{\sin nx}{n}\Big|_0^{\pi} - \int_0^{\pi}\frac{\sin nx}{n}\right]$$

$$= \frac{2}{\pi}\left[\frac{\pi \sin(\pi n)}{n} - 0\right] - \left[\cancel{\cos} - \frac{\cos nx}{n^2}\Big|_0^{\pi}\right]$$

$$\cancel{= \frac{2}{\pi}\left[0 - \left[\frac{-\cos \pi n}{n^2} - \frac{1}{n^2}\right]\right]}$$

$$= \frac{2}{n^2\pi}\left\{\cos n\pi - \cos 0\right\}$$

$$= \frac{2}{n^2\pi}\left\{(-1)^n - 1\right\}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\, dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} |x| \sin nx\, dx = 0 \;; \quad \because \int_{-a}^{a} f = 0; \text{ If } f \text{ is an odd function.}$$

Even x Odd = 0.

$$\text{Eq } 1 \Rightarrow f(x) = \pi/2 + \sum_{n=1}^{\infty} \frac{2}{n^2\pi}\left\{(-1)^n - 1\right\}\cos nx + 0$$

Ans

Even pe 0, Odd pe -2.

$$= \frac{\pi}{2} + \frac{2}{\pi}\sum_{i=1}^{\infty} \frac{(-2)\cos(2n-1)x}{(2n-1)^2}$$

$$= \frac{\pi}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2}$$

13) $f(x) = \begin{cases} x & -\pi < x \leq 0 \\ \pi - x & 0 < x < \pi \end{cases}$ $\quad f(x+2\pi) = f(x)$

Required Fourier Series is:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0} x\,dx + \int_{0}^{\pi} (\pi - x)\,dx\right.$$

$$= \frac{1}{2\pi}\left[\frac{x^2}{2}\Big|_{-\pi}^{0} + \pi\left\{x\Big|_0^{\pi}\right\} - \frac{x^2}{2}\Big|_0^{\pi}\right\}$$

$$= \frac{1}{2\pi}\left[-\frac{\pi^2}{2} + \pi(\pi) - \frac{\pi^2}{2}\right]$$

$$= \frac{1}{2\pi}[0] = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} x\cos nx\,dx\right.$$

$$+ \int_0^{\pi} (\pi - x)\cos nx\,dx$$

$$= \frac{1}{\pi}\left[\left\{\frac{x\sin nx}{n}\Big|_{-\pi}^{0} - \int_{-\pi}^{0} \frac{(1)\sin nx}{n}\right\} + \pi\left\{\frac{\sin nx}{n}\Big|\right.\right.$$

$$- \left\{\frac{x\sin nx}{n}\Big|_0^{\pi} - \int_0^{\pi} \frac{(1)\sin nx}{n}\right\}\Bigg]$$

$\sin n0 = \sin n\pi = 0$

$$= \frac{1}{\pi}\left[\frac{\cos nx}{n^2}\Big|_{-\pi}^{0} - \frac{\cos nx}{n^2}\Big|_0^{\pi}\right]$$

$$= \frac{1}{n^2\pi}\left[\{\cos 0 - \cos n(-\pi)\} - \{\cos n\pi - \cos 0\}\right]$$

$$= \frac{1}{n^2\pi}\left[1 - (-1)^n - ((-1)^n - 1)\right]$$

$$= \frac{1}{n^2\pi}\left[2 - 2(-1)^n\right]$$

$$= \frac{2[1 - (-1)^n]}{n^2\pi}$$

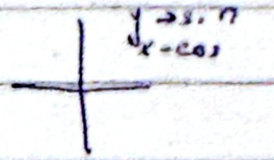
Calc. b_n.

19)

$a_0 = \frac{1}{2\pi}\int_0 f(x)\,dx$ —> cannot use evenfunction property since nonegative limit.

$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx$

$= \frac{1}{\pi}\int_0^{\pi} x^2 \cos nx\,dx$



$\cos n\pi = (-1)^n$; $\sin n\pi = 0$.

$\sin \frac{n\pi}{2}$

$\sin\frac{\pi}{2} = 1$, $\sin\frac{2\pi}{2} = 0$ 1, 0, -1, 0, 1, 0, -1, 0

2, 4, 6, 8, ...

$T_n = 2n \to 0$

$\sin\frac{3\pi}{2} = -1$, $\sin\frac{4\pi}{2} = 0$

1, 3, 5, 7, ...

$\sin\frac{5\pi}{2} = 1$, $\sin\frac{6\pi}{2} = 0$

$T_n = 2n-1$

$(2n-1)(-1)^{n+1}$

$\sin\frac{7\pi}{2} = -1$, $\sin\frac{8\pi}{2} = 0$

20) $f(x) = \begin{cases} -\pi/2 \;; & -\pi < x < -\pi/2 \\ x \;; & -\pi/2 < x < \pi/2 \\ \pi/2 \;; & \pi/2 < x < \pi \end{cases}$

Req. F. Series is

$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n$

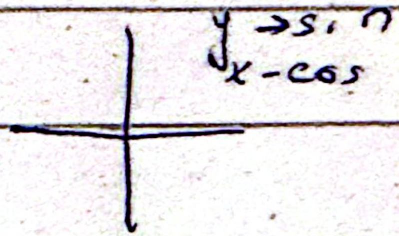
$\frac{1}{\pi}\left\{ \int_{-\pi}^{-\pi/2} -\frac{\pi}{2}\,dx + \int_{-\pi/2}^{\pi/2} x\,dx + \int_{\pi/2}^{\pi} \frac{\pi}{2}\,dx \right\}$

15) has a formula diff due to limits.

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\, dx$$ ——> cannot use even function property since no negative limit.

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\, dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x^2 \cos nx\, dx$$



$\cos n\pi = (-1)^n$; $\sin n\pi = 0$.

$\sin \frac{n\pi}{2}$

$\sin 2\pi$ 0 $\quad$ 1, 0, -1, 0, 1, 0, -1, 0

...even function
, ...rty since nonegative
limit.

$$\frac{\ }{\pi}\int_0^{\ } \ldots x^2 \cos nx \, dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x^2 \cos nx \, dx$$

$\int$ → sin, x - cos

$\cos n\pi = (-1)^n$; $\sin n\pi = 0$.

$\sin \frac{n\pi}{2}$

$\sin \frac{\pi}{2} = 1$, $\sin \frac{2\pi}{2} = 0$ — $1, 0, -1, 0, 1, 0, -1, 0$

$2, 4, 6, 8, \ldots$

$\sin \frac{3\pi}{2} = -1$, $\sin \frac{4\pi}{2} = 0$ — $T_n = 2n \to 0$

$1, 3, 5, 7, \ldots$

$\sin \frac{5\pi}{2} = 1$, $\sin \frac{6\pi}{2} = 0$ — $T_n = 2n-1$

$(2n-1)(-1)^{n+1}$

$\sin \frac{7\pi}{2} = -1$, $\sin \frac{8\pi}{2} = 0$

20) $f(x) = \begin{cases} -\pi/2 ; & -\pi < x < -\pi/2 \\ x ; & -\pi/2 < x < \pi/2 \\ \pi/2 ; & \pi/2 < x < \pi \end{cases}$

Req. f. Series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, dx = \frac{1}{2\pi}\left[\int_{-\pi}^{-\pi/2} -\frac{\pi}{2} dx + \int_{-\pi/2}^{\pi/2} x \, dx + \int_{\pi/2}^{\pi} \frac{\pi}{2} dx\right]$$

$$a_0 = \frac{1}{2\pi}\left[-\frac{\pi}{2}\int_{-\pi}^{-\pi/2} dx + \underbrace{0}_{\text{(due to odd fn)}} + \frac{\pi}{2}\int_{\pi/2}^{\pi} dx\right]$$

$$= \frac{1}{2\pi} \left\{ -\frac{\pi}{2} \cdot x \Big|_{-\pi}^{-\pi/2} + \frac{\pi}{2} \; x \Big|_{\pi/2}^{\pi} \right.$$

$$= \frac{1}{2\pi} \left\{ -\frac{\pi}{2} \left(-\frac{\pi}{2} + \pi \right) + \frac{\pi}{2} \left(\pi - \frac{\pi}{2} \right) \right\}$$

$$= \frac{1}{2\pi} \cdot \frac{\pi}{2} \left\{ \cancel{\frac{\pi}{2}} - \cancel{\pi} + \cancel{\pi} - \cancel{\frac{\pi}{2}} \right\}$$

$$= 0.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x \cdot \cos nx = 0.$$

Odd $\times$ Even = Odd function

$$\int_{-a}^{a} f = 0 \text{ if } f \text{ is odd}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin x$$

odd $\times$ odd = even

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx = \frac{1}{\pi} 2 \int_{0}^{\pi} f(x) \sin nx \, dx$$

$$= 2 \left[\int_{0}^{\pi/2} x \sin nx \, dx + \int_{\pi/2}^{\pi} \frac{\pi}{2} \sin nx \, dx \right.$$

$$\int uv = u \int v - \int u' \int v$$

$$= \frac{2}{\pi}\left[\left\{ \frac{x(-\cos nx)}{n}\Big|_0^{\pi/2} - \int_0^{\pi/2} \frac{(1)\cos nx\,dx}{n} \right\} + \frac{\pi}{2}\int_{\pi/2}^{\pi} \sin nx\,dx \right]$$

$$= \frac{2}{\pi}\left[\left\{ \frac{-\frac{\pi}{2}\cos\frac{n\pi}{2} - 0}{n} \right\} + \frac{\sin nx}{n^2}\Big|_0^{\pi/2} + \frac{\pi}{2}\,\frac{(-\cos nx)}{n}\Big|_{\pi/2}^{\pi} \right]$$

$$= \frac{2}{\pi}\left[\frac{-\pi\cos\frac{n\pi}{2}}{2n} + \frac{1}{n^2}\left(\sin\frac{n\pi}{2}\right) - \frac{\pi}{2n}\left\{ \cos n\pi - \cos\frac{n\pi}{2} \right\} \right]$$

$$= \frac{2}{\pi}\left\{ \frac{\sin\frac{n\pi}{2}}{n^2} - \frac{\pi}{2n}(-1)^n \right\}$$

Required fourier Series is:

$$f(x) = 0 + \sum_{n=1}^{\infty} 0 + \frac{2}{\pi}\left\{ \frac{\sin\frac{n\pi}{2}}{n^2} - \frac{\pi}{2n}(-1)^n \right\}\sin nx$$

Ans

$$= \frac{2}{\pi}\left\{ \sum_{n=1}^{\infty} \frac{\sin\frac{n\pi}{2}}{n^2}\sin nx - \frac{\pi}{2}\sum_{n=1}^{\infty} \frac{(-1)^n \sin nx}{n} \right\}$$

$$\boxed{\sin(2n-1)\frac{\pi}{2}}$$

$$= \frac{2}{\pi}\left\{ \sum_{n=1}^{\infty} \frac{\sin(2n-1)\frac{\pi}{2}}{(2n-1)^2}\sin(2n-1)x - \frac{\pi}{2}\sum_{n=1}^{\infty} \frac{(-1)^n \sin nx}{n} \right\}$$

1, -1, 1, -1, 1, -1, ...

$$\frac{2}{\pi}\left\{ \sum_{n=1}^{\infty} \frac{(-1)^{n+1}\sin(2n-1)x}{(2n-1)^2} - \frac{\pi}{2}\sum_{n=1}^{\infty} \frac{(-1)^n \sin nx}{n} \right\}$$

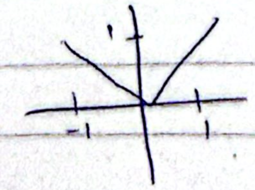
* If $f(x)$ is periodic function of arbitrary period '$2L$'

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}$$

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx \; ; \; a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\,\sin\frac{n\pi x}{L}\,dx$$

Ex. 11.2 Q8-17



$f(x) = |x|$; $-1<x<1$ or $f(x) = \begin{cases} -x & -1<x\leq 0 \\ x & 0<x<1 \end{cases}$

$2L = 2$

$L = 1$

$$a_0 = \frac{1}{2L}\int_{-2}^{2} f(x)\,dx = \frac{1}{2}\int_{-1}^{1}|x|\,dx = \frac{1}{\cancel{2}}\cancel{2}\int_{0}^{1}x\,dx$$

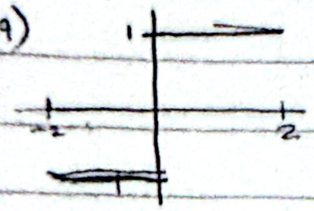
$$= \frac{x^2}{2}\Big|_0^1 = \frac{1}{2}.$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\,\sin\frac{n\pi x}{1} = \frac{1}{1}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L} = \frac{1}{1}\int_{-1}^{1} |x| \cos\frac{n\pi x}{1}\, dx$$

$$2\int_{0}^{-1} x \cos n x \, dx$$

Ex# 11.2 Q 8-17

9) 

$2L = 4$

$\Rightarrow L = 2$

$$f(x) = \begin{cases} -1 ; & -2 < x < 0 \\ 1 ; & 0 < x < 2 \end{cases}$$

$-L$ represents lower limit of $f(n)$ not literal $-L$.

$f(-1.3) = -1$

$f(1.3) = 1$

$f(-x) = -f(x)$ - therefore this function is odd.

Req. Fourier Series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}$$

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\cdot dx.$$

$$= \frac{1}{4}\Big\{ 0 \Big\}_{-2}^{2} = 0.$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$$

$$= \frac{1}{2}\int_{-2}^{2} f(x)\cos\frac{n\pi x}{2}\,dx = 0.$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{2}\,dx$$

$$= \frac{1}{2}\int_{-2}^{2} f(x)\sin\frac{n\pi x}{2}\,dx$$

$$= \frac{1}{\cancel{2}}(\cancel{2})\int_{0}^{2} f(x)\sin\frac{n\pi x}{2}$$

$$= \int_{0}^{2}(1)\sin\frac{n\pi x}{2} = \left.\frac{-\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right|_{0}^{2}$$

$$= -\frac{2}{n\pi}\left\{\cos n\pi - \cos 0\right\}.$$

$$b_n = \frac{-2}{n\pi}\left\{(-1)^n - 1\right\}; \quad \because \cos n\pi = (-1)^n$$

Eq $1 \Rightarrow f(x) = 0 + \sum_{n=1}^{\infty}(0)\cos\frac{n\pi x}{2} - \frac{2}{n\pi}\left\{(-1)^n - 1\right\}$

$$f(x) = \frac{-2}{\pi}\sum_{n=1}^{\infty}\frac{\{(-1)^n - 1\}}{n}\sin\frac{n\pi x}{2}$$

OR

$$\begin{cases} -1 \; ; & -2 < x < 0 \\ 1 \; ; & 0 < x < 2 \end{cases} = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\sin\frac{(2n-1)\pi x}{2}}{(2n-1)}$$

Half Range Series — One of the coefficients is zero

Odd – Sin series   Even – cos.

$z = x + iy$ or $a + ib$ or $r(\cos\theta + i\sin\theta)$ or $re^{i\theta}$

Rectangular / Cartesian form

Polar form.

$x = r\cos\theta,\ y = r\sin\theta$

$\theta = \tan^{-1}(y/x)$

$r = \sqrt{x^2 + y^2}$

Ch. 16.2

Complex valued function / Complex function

these are the function that are defined over a complex domain.

$$f(\overset{\text{complex}}{z}) = u(x,y) + i\,v(x,y)$$

$$f(z) = z^2 = (x+iy)^2 = \underbrace{(x^2 - y^2)}_{u} + i\underbrace{2xy}_{v}$$

Zeroes and poles :-

Let $f(z)$ is a complex valued function then (i) $z = z_0$ is called 'zero' of this function. If $f(z_0) = 0$

(ii) $z = z_0$ is called 'pole' of this function if $f(z_0) = \infty$

Example : $f(z) = \dfrac{(z-2)(z-1)^2}{(z-3)(z-4)}$

Find all zeroes and poles of the above function.

For zeros:

$f(z_0) = 0$

$\dfrac{(z-2)(z-1)^2}{(z-3)(z-4)} = 0$

$(z-2)(z-1)^2 = 0$

~~$z=2$ or $z=1$~~

$z = 1, 2.$

For poles :-

$f(z) = \infty$

$\dfrac{(z-2)(z-1)^2}{(z-3)(z-4)} = \infty$

$= \dfrac{(z-2)(z-1)^2}{(z-3)(z-4)} = \dfrac{1}{0}$

$= (z-3)(z-4) = 0$

$z = 3, 4$

Section 13.4

Cauchy Rieman equations.

If $f(z) = u(x,y) + iv(x,y) = U + iv$ is a complex valued function then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

C-R equations

These functions (that satisfy C-R eq.) are called analytic functions

$f(z) = z^2 = (x+iy)^2 = (x^2-y^2) + i2xy$

Here

$U = x^2-y^2, \quad v = 2xy$

$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = -2y$

$\frac{\partial v}{\partial x} = 2y, \quad \frac{\partial v}{\partial y} = 2x$

From above calculations.

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$2x = 2x \qquad 2y = -2y$

$\because$ C-R equations are satisfied

$\therefore$ $f(z)$ is an analytic function.